
What Shapes Student CGPA?

An Engineering Student Academic Performance Dataset and Machine Learning Analysis

CSE 4112: Machine Learning Laboratory

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14 September 2026

Abstract

This course report documents a questionnaire dataset collected from undergraduate engineering students in Bangladesh. Two independently administered Google Forms captured the same core academic, behavioural, lifestyle, motivation, support, and study-environment concepts with small differences in wording, language, ordering, and grade-band options. The source files contain 1,126 responses: 230 from BUET, BRAC University, and CUET, and 896 from KUET. A reproducible pipeline maps both forms to a common schema, removes source-specific text differences, normalises university and department names, reconciles grade bands, records missing-value decisions, and removes 18 records without a usable CGPA target. The resulting dataset contains 1,108 cleaned student records from four universities and represents both recent SGPA and cumulative CGPA as four ordered groups from C0 to C3. The project package includes raw and cleaned tabular files, WEKA-ready ARFF files, a data dictionary, cleaning logs, descriptive figures, fixed train/test indices, model-output tables, and executable notebooks. These resources support reuse in educational data analysis, questionnaire harmonisation, ordinal classification, interpretable modelling, and comparison of standard machine-learning algorithms. The latter part of this report demonstrates machine-learning analyses that can be performed on the dataset while keeping data description, predictive evaluation, and interpretation as separate tasks.

Keywords: academic performance; questionnaire data; engineering students; educational analytics; ordinal classification; interpretable machine learning; university survey

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Part I: Data Article Section

1 Specifications table

Table 1: Data in Brief-inspired dataset specifications.

Subject	Computer Science and Educational Data Science
Specific subject area	Student academic-performance survey data, questionnaire harmonisation, and educational machine learning
Type of data	Raw questionnaire responses; cleaned categorical and ordinal tables; CSV and ARFF files; figures; model-output tables; executable notebooks
Data collection	Two bilingual Google Forms distributed to engineering students at BUET, BRAC University, CUET, and KUET. The forms used categorical answer options and related but not identical wording.
Data source location	Bangladesh; undergraduate engineering students from four universities
Data accessibility	Course project repository. TODO: insert the confirmed public repository URL if the team publishes the package.
Related research article	None. This document is a Data in Brief-inspired course report followed by an ML analysis.

2 Value of the Data

- The dataset records academic history, study habits, lifestyle, motivation, support, and study environment for engineering students using one harmonised schema.
- Responses from four universities and two separately administered questionnaires support analysis of both pooled and source-specific distributions.
- The preserved raw-to-clean mapping and cleaning log allow researchers to inspect how bilingual response labels, free-text departments, and two grade-band schemes were reconciled.
- Ordered CGPA and SGPA groups support reproducible classification, ordinal-error analysis, and interpretable Decision Tree experiments.
- The package provides both Python and WEKA formats, making it useful for teaching model comparison, validation, and confusion-matrix interpretation.
- Researchers can reuse the data to study questionnaire design and to test whether group-level associations translate into predictions for unseen students.

3 Background

Academic performance reflects both prior preparation and the context in which students study. Educational data mining can organise these observations into measurable variables, compare student groups, and test whether the measured information supports prediction. Previous work has used classical classifiers, ensembles, and explainability methods for student-performance tasks [1, 2, 3]. Public Bangladeshi datasets also show the value of combining academic records with attendance, study time, technology use, family context, and lifestyle indicators [4, 5, 6, 7].

The present project focuses on engineering students at BUET, BRAC University, CUET, and KUET. It extends an initial multi-university questionnaire with a larger KUET questionnaire that asks the same substantive questions with some different wording and response formatting. The project therefore has two connected aims. The first is to create a transparent, reusable dataset whose origin, response options, cleaning decisions, and limitations can be traced. The second is to demonstrate how that dataset can support descriptive analysis and machine-learning experiments without treating prediction as causal evidence.



Figure 1: Project workflow from questionnaire collection to interpretable evaluation.

4 Data description

4.1 Dataset overview

The two source forms contain 1,126 responses. Target cleaning removes 18 rows without a usable CGPA response, giving 1,108 final records. Table 2 reports the institutional composition.

Table 2: Dataset composition after target cleaning.

University	Records	Share (%)
KUET	878	79.2
BUET	104	9.4
BRAC University	101	9.1
CUET	25	2.3
Total	1,108	100.0

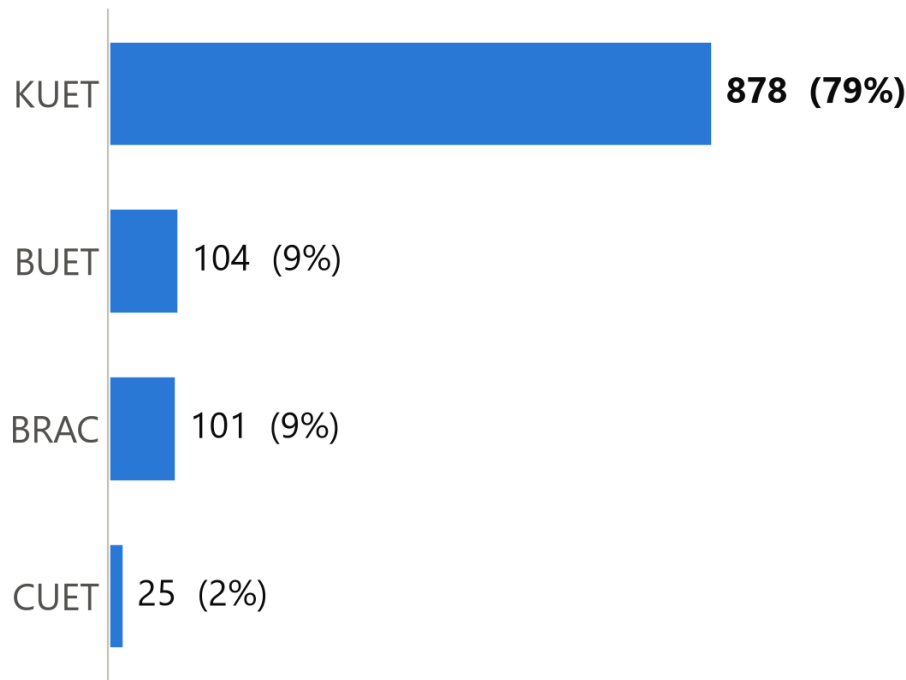


Figure 2: University distribution in the cleaned dataset.

4.2 Questionnaire structure

The common questionnaire covers academic context, study habits, lifestyle, pressure, motivation, support, environment, and grade outcomes. The two forms ask the same core questions but do not always use identical English text or answer formatting. Figure 3 shows representative form content. No names or roll numbers appear in the modelling table.

DATA COLLECTION

What Did the Form Look Like?

University Student Performance Analysis Form

This questionnaire collects behavioral, demographic, and socioeconomic indicators to understand factors affecting academic performance. This survey is completely anonymous; no names, roll numbers, or exact income figures are collected.

(শিক্ষাগত ফলাফলের ওপর প্রভাব বিচারকল্পী বিষয়গুলো বোঝার লক্ষ্যে এই প্রশ্নমালায় মাধ্যমে আচরণগত, জনমিত্তিক এবং আর্থ-সামাজিক সূচকসমূহ সংগ্রহ করা হচ্ছে। এই জরিপটি সম্পূর্ণ বেনামী। এতে কোনো নাম, রোল নম্বর বা আয়ের সুনির্দিষ্ট পরিমাপ সংগ্রহ করা হয় না।)

আপনার উত্তর শুধুমাত্র গুগল-এ সাইন-ইন করুন। আরও জানুন

* প্রয়োজনীয় প্রশ্ন নির্দেশ করে

What is the name your university?
(তোমার বিশ্ববিদ্যালয়ের নাম কি?)

আপনার উত্তর

Did you get into your desired subject or department?
(তুমি কি তোমার পছন্দের সাজেশ্ব/ডিপার্টমেন্টে ভর্তি হতে পেরেছো?)

☐ Yes, exactly what I wanted

☐ Close to what I wanted

☐ Somewhat different

☐ Mostly different

☐ Not at all what I wanted

How often did your phone, gaming, social media, or entertainment steal your planned study time?
(তোমার ফোন, গেমিং, সোশ্যাল মিডিয়া বা বিনোদন কত ঘন ঘন তোমার পরিকল্পিত পড়ার সময় চুরি করে?)

☐ Never

☐ Rarely

☐ Sometimes

☐ Often

☐ Almost every day

How much time did tuition, a job, club activities or other responsibilities take each week?
(টিউশন, জব/ক্লাব/অন্যান্য দায়িত্ব প্রতি সপ্তাহে তোমার কতটা সময় ব্যয় করে?)

☐ None

☐ 1-5 hrs

☐ 6-10 hrs

☐ 11-15 hrs

☐ More than 15 hrs

When academic life became difficult, how much support did you receive from family or close friends?
(শিক্ষাজীবনের কঠিন সময়ে পরিবার থেকে তুমি কতটুকু সহায়তা পেছো?)

☐ No support

☐ Very little support

☐ Some support

☐ Good support

☐ Strong support

Real screenshots of the University Student Performance Analysis Form — English + Bangla, anonymous.
The KUET form (896 responses) asked the same core questions with slightly different wording; both were merged.

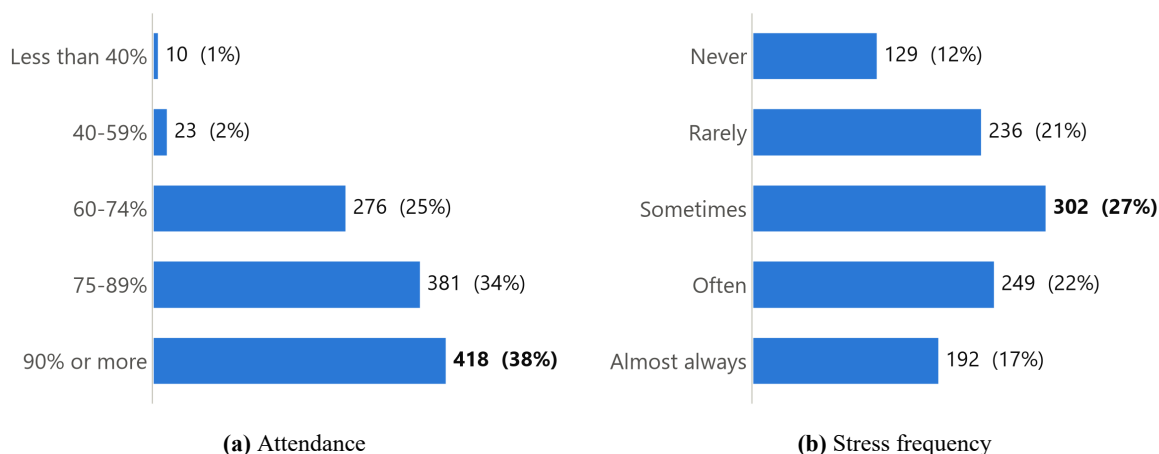
Figure 3: Representative form questions from the multi-university questionnaire. The KUET form used the same core concepts with some different wording.

Table 3: Questionnaire variable groups.

Group	Variables	Typical response form
Academic	University, department, semester, recent SGPA, CGPA	Nominal fields and ordered bands
Study habits	Attendance, weekly study time, study style, topic clarity	Five ordered categories
Lifestyle	Sleep duration, phone or social-media distraction	Five ordered categories
Pressure	Stress frequency, outside commitments	Frequency or hour bands
Motivation and support	Desired department, admission satisfaction, career expectation, family or friend support	Five ordered categories
Environment	Study-friendly living place, routine manageability	Five ordered categories

4.3 Response distributions

All descriptive counts use the complete 1,108-row cleaned snapshot. For example, 418 students (37.7%) report attendance of 90% or more, 302 (27.3%) report that stress sometimes blocks study, and 310 (28.0%) report five to six hours of sleep. These figures describe answer frequencies only.

**Figure 4:** Examples of cleaned questionnaire response distributions.

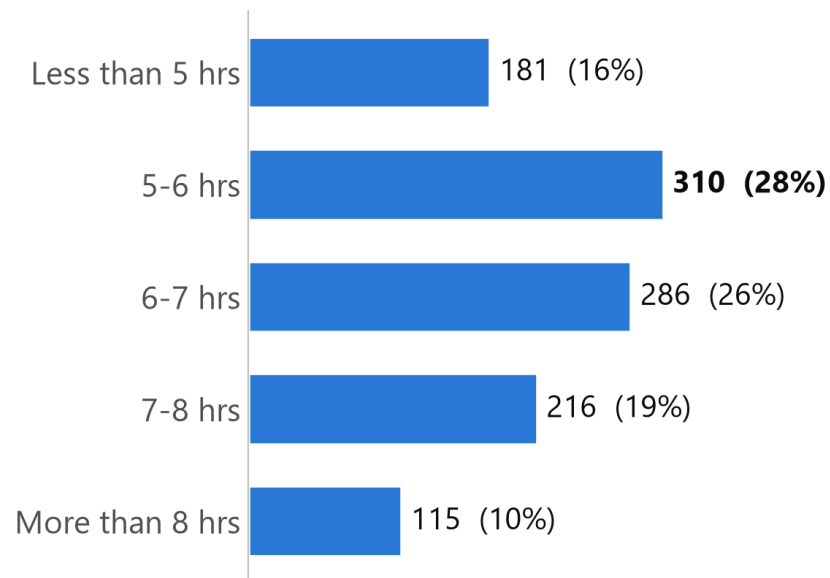


Figure 5: Reported sleep duration in the cleaned dataset.

4.4 Target classes

Both forms record grade ranges rather than a consistent exact numeric CGPA. The common target is therefore an ordered four-class variable.

Table 4: CGPA target groups.

Code	CGPA range	Count	Share (%)
C0	Below 3.20	236	21.3
C1	3.20–3.49	321	29.0
C2	3.50–3.74	303	27.3
C3	3.75 and above	248	22.4

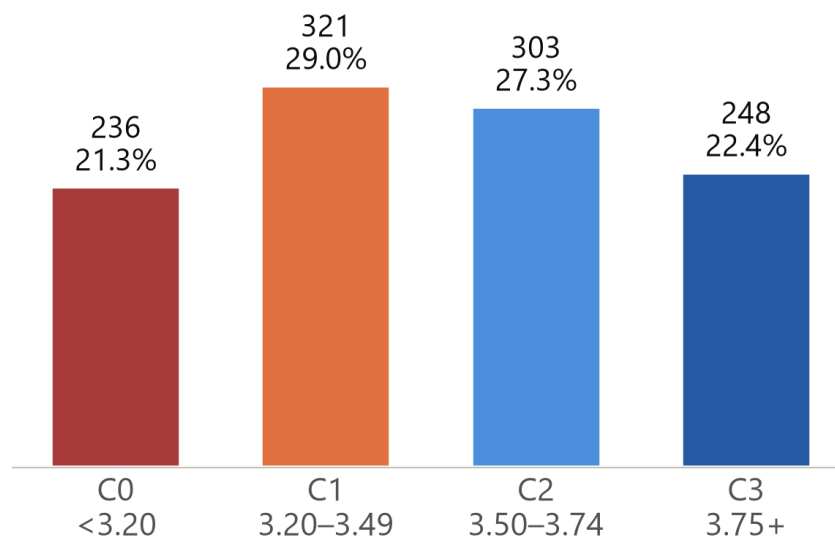


Figure 6: Distribution of the four CGPA target groups.

4.5 Reference dataset comparison

The IUBAT reference dataset contains 894 Computer Science students and 31 columns, including exact numeric CGPA and SGPA, attendance, study hours, social-media hours, co-curricular activity, and socioeconomic variables [4]. The current questionnaire uses grade bands and adds direct questions about stress, sleep, topic clarity, desired department, support, study environment, and routine manageability.

IUBAT_Student_Performance_Dataset.csv (classified_dataset.csv adds Performance_Category)

	A	B	C	D	E	F	G	H
1	Current Semester	How many hour do you study daily? (Hours)	How many hour do you spent daily in social media? (Hours)	Average attendance on class (Percentage)	Are you engaged with any co-curriculum activities?	What was your previous SGPA?	What is your current CGPA?	Performance_Category
2	4	2	2	100	No	3.8	3.64	Good Performance
3	4	3	2	90	No	3.4	3.53	Good Performance
4	4	2	1	95	Yes	3.93	3.89	Good Performance
5	4	1	3	70	No	3.1	3.5	Poor Performance
6	4	3	2	96	Yes	3.81	3.65	Good Performance
7	4	1	1	85	Yes	3.89	3.81	Good Performance
8	4	3	1	97	No	3.79	3.82	Good Performance
9	4	2	2	95	Yes	3.28	3.32	Good Performance
10	4	2	3	99	No	3.78	3.88	Good Performance
11	2	3	2	100	No	3.9	3.9	Good Performance

Rendered preview of the first 10 rows of the reference dataset (Prima 2024, Mendeley Data).

Figure 7: Preview of the IUBAT reference dataset used to compare feature coverage.

4.6 Files and reproducibility

The project package contains raw response files, three cleaned merged tables, WEKA-ready CSV and ARFF files, executed notebooks, machine-readable result summaries, figures, a data dictionary, cleaning logs, fixed split indices, and WEKA output logs. Figure 8 illustrates the cleaned row structure.

merged_full_clean.csv — one row = one student

	A	B	C	D	E	F	G	H
1	university	attendance	weekly_study_time	stress_frequency	weekly_responsibilities	desired_department_match	recent_sgpa_class	cgpa_class
2	KUET	75-89%	7-10 hrs	Almost always	More than 15 hrs	Not at all what I wanted	C0	C0
3	KUET	75-89%	More than 15 hrs	Never	11-15 hrs	Yes, exactly what I wanted	C1	C1
4	BRAC	90% or more	More than 15 hrs	Sometimes	More than 15 hrs	Yes, exactly what I wanted	C1	C2
5	KUET	75-89%	3-6 hrs	Sometimes	More than 15 hrs	Mostly different	C1	C1
6	KUET	60-74%	7-10 hrs	Rarely	6-10 hrs	Close to what I wanted	C1	C3
7	KUET	60-74%	3-6 hrs	Often	11-15 hrs	Yes, exactly what I wanted	C3	C0
8	KUET	75-89%	3-6 hrs	Sometimes	11-15 hrs	Not at all what I wanted	C2	C3
9	BUET	90% or more	3-6 hrs	Often	1-5 hrs	Mostly different	C1	C0
10	KUET	75-89%	11-15 hrs	Often	1-5 hrs	Not at all what I wanted	C3	C3
11	KUET	60-74%	11-15 hrs	Almost always	6-10 hrs	Yes, exactly what I wanted	C1	C1

Rendered preview of 10 cleaned rows (selected columns). No names, emails or IDs are collected.

Figure 8: Cleaned dataset preview. One row represents one student response.

5 Experimental design, materials, and methods

5.1 Questionnaire design and collection

Two bilingual Google Forms collected responses from undergraduate engineering students. Source A contains 230 responses from BUET, BRAC University, and CUET. Source B contains 896 KUET responses. Both instruments cover 19 substantive concepts, while their column order, English phrasing, bilingual answer formatting, and grade-band options differ. The collection windows were 31 July to 13 August 2026 for the KUET form and 22 August to 3 September 2026 for the multi-university form.

5.2 Schema mapping and harmonisation

Each raw column was mapped by hand to a common snake_case schema and guarded by assertions. Manual mapping was selected because automatic text similarity could incorrectly merge conceptually different satisfaction questions. The harmonisation pass removed emoji and Bangla parenthetical translations from answer values while preserving English department abbreviations. One study-style label was reconciled, and the two grade-band schemes plus typed KUET values were mapped into C0–C3.

The pipeline retained a source-form field for audit comparisons but excluded it from modelling. It dropped timestamps because the collection windows identify the source form, and it removed an empty email column. University spelling variants were reduced to four institutions. A rule-based canonicaliser reduced 110 free-text department strings to 24 department names, then grouped categories with fewer than 15 students for modelling.

5.3 Cleaning sequence

Table 5: Auditable cleaning sequence.

Step	Decision	Reason	Rows
1	Drop timestamp	Collection timing is a source proxy, not a student attribute	1,126
2	Normalise 10 university labels to 4	Case and suffix variants identify the same institution	1,126
3	Canonicalise 110 department strings to 24	Abbreviations, spelling, and free text otherwise split categories	1,126
4	Reconcile one study-style label	Keep the ordered scale consistent across forms	1,126
5	Build four CGPA groups	Four classes are recoverable from both questionnaires	1,126
6	Remove unusable CGPA targets	A supervised model cannot learn from an unlabelled target	1,108
7	Map supported commitment blanks to None	The KUET form provides an explicit None option with a comparable response rate	1,108
8	Fill remaining sparse predictor blanks	Within-source modal categories preserve the observed category labels	1,108
9	Encode ordered scales	Preserve the natural order of the responses	1,108
10	Derive academic progress and audit features	Replace raw semester with a comparable progress measure	1,108

5.4 Missing values and leakage control

Only 18 rows were removed, all because their CGPA target could not be interpreted. Blank outside-commitment answers were handled as None when the form design and response rates supported that meaning. The remaining predictor missingness was below 1.3% per field and was filled with the modal category from the same source form. Six rows had recent SGPA filled using information from their CGPA class in the earlier cleaning process. Descriptive summaries retain those rows, but every final model experiment excludes them. The resulting modelling set contains 1,102 students.

Result satisfaction was omitted from predictive feature sets because it follows an academic outcome. Source form was also excluded so the model could not learn questionnaire identity. University, department, and raw semester were not used in the 12-question teacher-facing trees.

5.5 Data quality checks

Quality checks covered class balance, source-form option overlap, university imbalance, duplicates, category coverage, missingness, and complete-case sensitivity. In the 1,108-row descriptive snapshot, target shares range from 21.3% to 29.0%. After the six-row modelling exclusion, the CGPA ZeroR baseline is 29.04% in 10-fold cross-validation. KUET contributes 79.2% of cleaned records and therefore requires an explicit generalisation warning. Complete-case recalculation changed no reported Spearman coefficient by more than 0.0036.

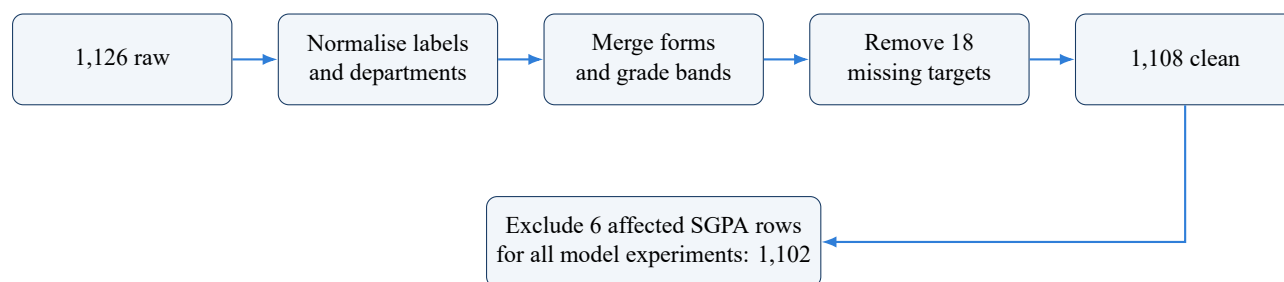


Figure 9: Cleaning and teacher-facing experiment populations.

6 Data limitations

- Grades and behaviours are self-reported and were not checked against registrar records.
- The convenience sample is dominated by KUET, which contributes 79.2% of the records.
- The survey is cross-sectional. It measures each student once and cannot observe changes across semesters.
- CGPA and SGPA are stored as bands rather than consistent exact numbers.
- Small wording and option differences remain between the two questionnaires after semantic harmonisation.
- Source A asks for CGPA before the most recent semester, while Source B asks for current CGPA. This one-semester reference difference affects interpretation of the SGPA feature.
- The dataset does not contain verified course-level grades, longitudinal activity, or equally sized samples from all universities.
- Questionnaire missing-value modes were calculated before cross-validation and before the fixed train/test split. This historical preprocessing order can transfer limited distribution information into evaluation data.

These limitations restrict population-level generalisation and any intended high-stakes use. The model scores are exploratory course-project evidence. They remain useful for reproducible classroom analysis when the sampling, measurement, and preprocessing boundaries stay visible.

Part II: Machine Learning Analysis

7 Machine-learning analysis framework

7.1 Research questions

1. Which questionnaire variables show the clearest descriptive relationship with CGPA groups?
2. Can the 12 questionnaire variables predict recent SGPA or CGPA for an unseen student?
3. How much predictive structure appears when recent SGPA is added to the CGPA inputs?
4. Which rules appear in a readable WEKA J48 tree, and how does that tree behave on the fixed test set?

7.2 Feature sets and targets

Three separate four-class tasks were prepared. Recent SGPA prediction uses 12 questionnaire variables: attendance, weekly study time, study style, topic clarity, sleep, stress, outside commitments, digital distraction, study environment, support, desired-department match, and routine manageability. The questionnaire-only CGPA task uses the same 12 inputs. The main CGPA task adds recent SGPA as a thirteenth input. The ordered classes are C0, C1, C2, and C3.

All descriptive summaries use 1,108 cleaned records. Every model uses 1,102 records after excluding six rows whose recent SGPA had previously been imputed using the CGPA class. University, department, semester, source form, identifiers, and result satisfaction are excluded from all three feature sets.

7.3 Two WEKA evaluation views

The model comparison uses stratified 10-fold cross-validation on all 1,102 eligible records with seed 1. J48, Random Forest, SMO, Logistic, Naive Bayes, and ZeroR are run on each of the three tasks under the same evaluation protocol.

The explanatory tree view uses the existing CGPA-stratified split with seed 42: 881 training rows and 221 test rows. The identical row indices are reused for all three targets. WEKA J48 uses confidence factor 0.25 and minimum leaf size 40 so that its exact tree is readable. The setting was specified for explanation rather than selected from test accuracy. ZeroR is fitted on the training rows and evaluated on the same test rows.

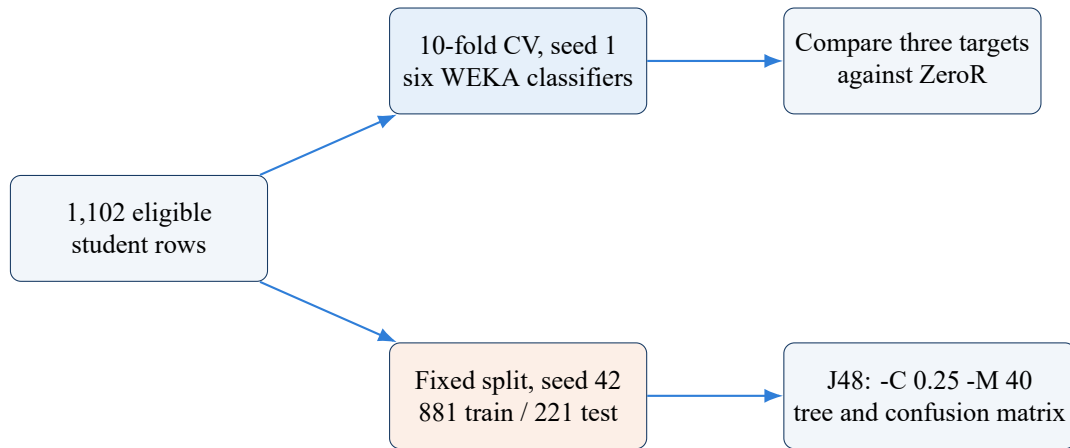


Figure 10: The WEKA model comparison and explanatory J48 evaluation use separately labelled protocols.

7.4 Preprocessing boundary

The raw-to-clean verification was re-executed before the final WEKA run and reproduced every modelling column used in the ARFF files. Questionnaire missing-value modes, however, were calculated in the historical cleaning pipeline before cross-validation and before the fixed split. This can transfer limited distribution information across folds. The reported scores are therefore exploratory course-project estimates rather than deployment estimates.

Table 6: Final WEKA prediction tasks.

Prediction target	Inputs	Purpose
Recent SGPA	12 questionnaire answers	Test the predictive signal for recent academic performance
CGPA questionnaire-only	12 questionnaire answers	Measure how far survey responses alone rise above the CGPA baseline
Main CGPA	12 answers plus recent SGPA	Measure the strongest available CGPA task and inspect its readable rules

8 Models and evaluation

8.1 Models

The final comparison uses six WEKA 3.8.7 classifiers [8]. J48 supplies a readable decision tree. Random Forest supplies a non-linear ensemble benchmark. SMO supplies the course-tool support-vector-machine implementation. Logistic Regression gives a linear classifier, Naive Bayes gives a probabilistic comparison, and ZeroR predicts only the majority class.

The 10-fold J48 run uses confidence factor 0.25 and WEKA’s default minimum leaf size of 2. Random Forest uses 100 trees and seed 1. The explanatory fixed-split J48 uses confidence factor 0.25 and minimum leaf size 40. This larger leaf size keeps the exact WEKA tree readable. The cross-validation J48

and the fixed-split J48 answer different questions and are reported separately.

8.2 Metrics

Table 7: Evaluation measures and interpretation.

Measure	Role
Exact accuracy	Primary fraction of students assigned to the correct group
Weighted F1	Frequency-weighted class performance reported by WEKA
Cohen's kappa	Agreement after accounting for chance
Confusion matrix	Counts for every actual and predicted group
Absolute class distance	Ordinal error magnitude: 0 exact, 1 adjacent, 2 or 3 farther away

The majority baseline depends on the target and evaluation subset. In 10-fold cross-validation, ZeroR reaches 26.86% for recent SGPA and 29.04% for both CGPA tasks. On the 221-row fixed test set, ZeroR reaches 27.15% for recent SGPA and 28.96% for both CGPA tasks. Scores close to those values require cautious interpretation.

9 Results

9.1 Descriptive CGPA patterns

Three questionnaire variables show the clearest corrected association with CGPA: desired-department match ($\rho = +0.119$, adjusted $q = 0.0010$), attendance ($\rho = +0.090$, $q = 0.0197$), and outside commitments ($\rho = -0.086$, $q = 0.0202$). Each association is small. Figures 11–13 show group composition rather than causal effects.

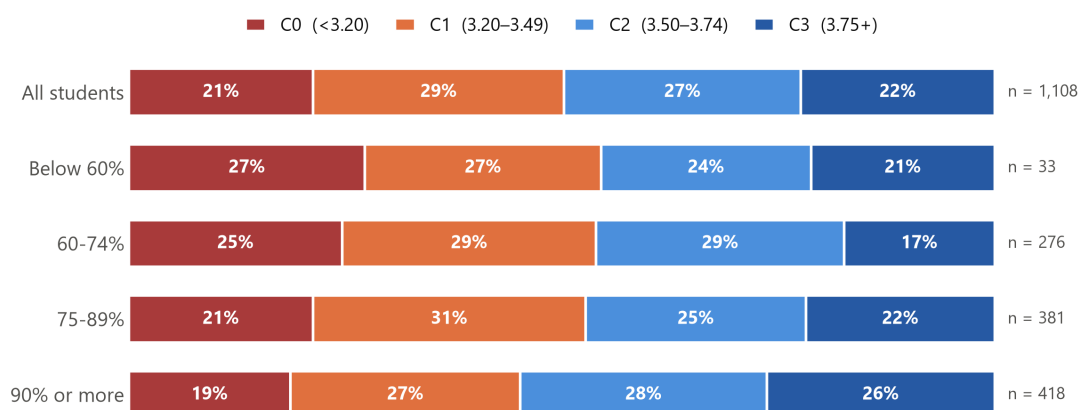


Figure 11: CGPA-group composition by attendance band. Students reporting 90% or more attendance have a 55% C2/C3 share, compared with 46% for the 60–74% group.

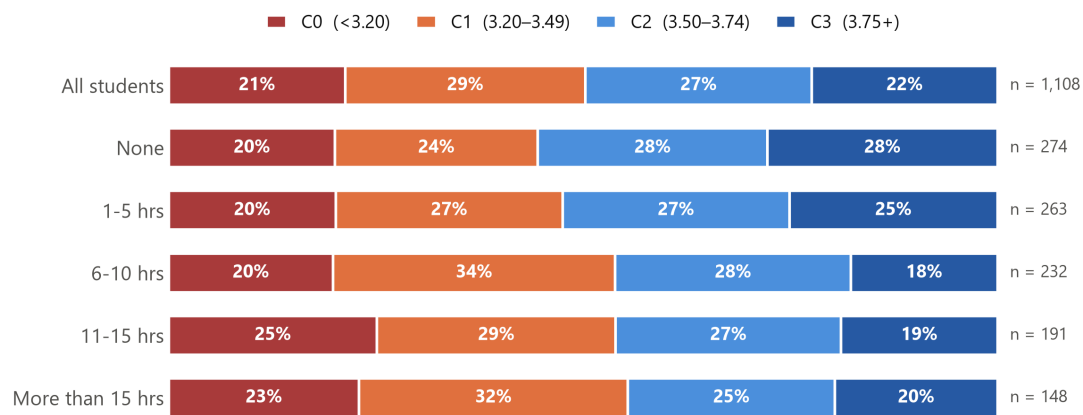


Figure 12: CGPA-group composition by weekly outside commitments. The C2/C3 share is 56% for no outside commitments and 45% above 15 hours.

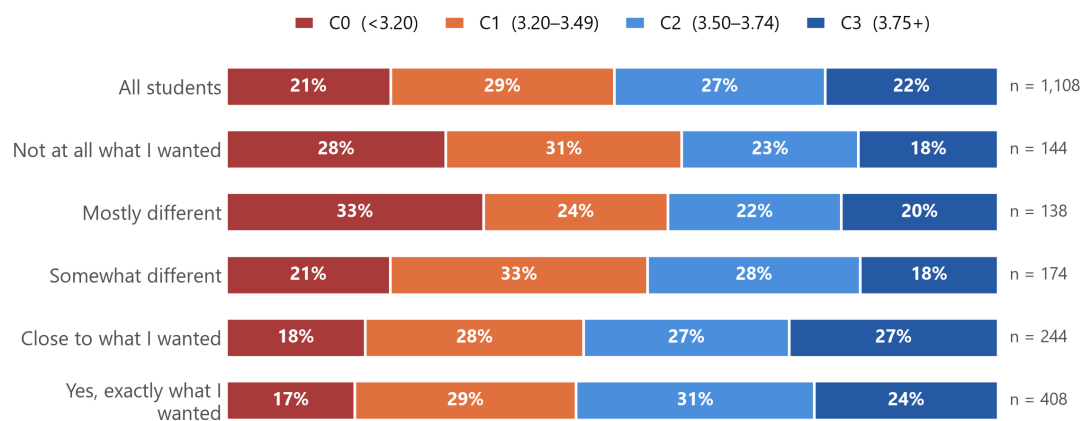


Figure 13: CGPA-group composition by desired-department match. The C2/C3 share is 55% among students who report exactly the department they wanted and 41% among those reporting no match.

9.2 WEKA 10-fold cross-validation

Table 8 reports the final like-for-like model comparison on all 1,102 eligible records. J48 is the strongest SGPA model at 30.13%, only 3.27 percentage points above ZeroR. Naïve Bayes is strongest for questionnaire-only CGPA at 30.04%, 1.00 point above ZeroR. Random Forest is strongest for the main CGPA task at 43.92%, 14.88 points above ZeroR. These results show weak prediction from questionnaire answers alone and a clearer predictive contribution from recent SGPA.

Table 8: WEKA stratified 10-fold cross-validation accuracy on 1,102 records, seed 1.

Model	Recent SGPA (%)	Questionnaire CGPA (%)	Main CGPA (%)
J48	30.13	26.68	38.29
Random Forest	29.76	29.40	43.92
SMO	25.95	27.86	37.11
Logistic	24.95	29.58	36.84
Naïve Bayes	26.68	30.04	38.11
ZeroR	26.86	29.04	29.04

WEKA 3.8.7 stratified 10-fold cross-validation (N = 1,102, seed = 1)

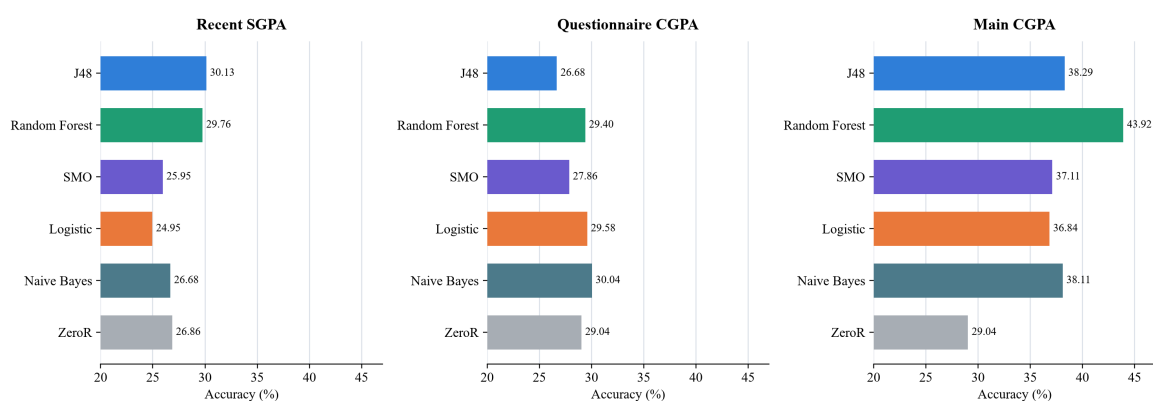


Figure 14: The final WEKA cross-validation comparison. Each group uses the same 1,102 eligible students and evaluation seed.

9.3 Readable WEKA J48 trees on the fixed split

The fixed-split J48 runs use the same 881 training and 221 test rows with `-C 0.25 -M 40`. The larger leaf requirement produces compact trees that can be shown without redrawing them. Table 9 reports each tree against the ZeroR test baseline.

Table 9: Fixed-split WEKA J48 results.

Prediction	Train (%)	Test (%)	Baseline (%)	Change (pp)
Recent SGPA	39.39	29.86	27.15	+2.71
Questionnaire CGPA	36.55	32.13	28.96	+3.17
Main CGPA	46.20	49.77	28.96	+20.81

The SGPA tree begins with routine manageability and creates 13 leaves. Its 29.86% test accuracy remains close to the 27.15% baseline, so its branches are exploratory. Figure 15 is the actual WEKA TreeVisual-izer export from this run.

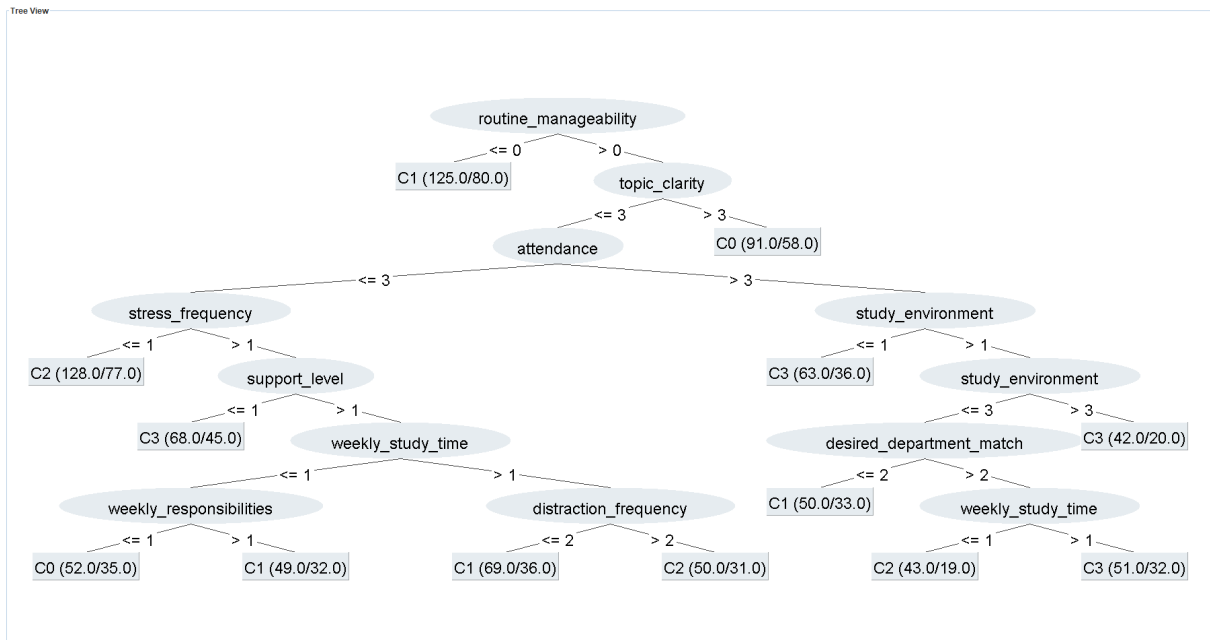


Figure 15: Actual WEKA J48 tree for recent SGPA from the 12 questionnaire inputs.

The questionnaire-only CGPA tree reaches 32.13% on the fixed test set, 3.17 points above its baseline, but the 10-fold model comparison remains near baseline. Figure 16 shows why the result should be treated cautiously: the tree uses several survey variables while the four classes remain mixed.

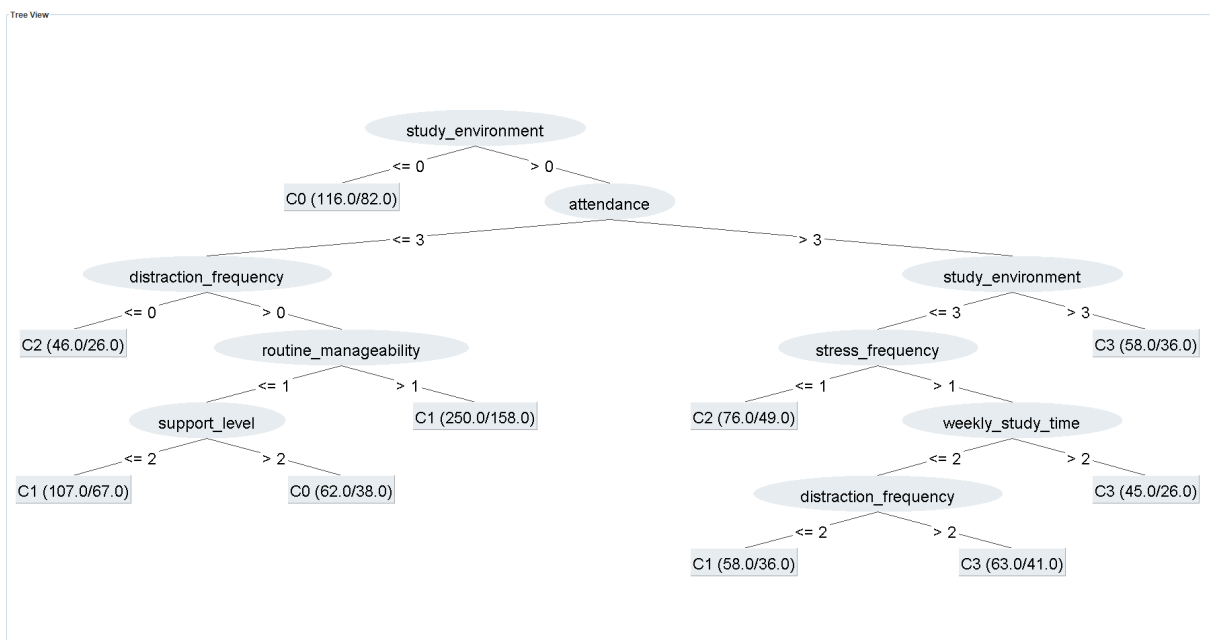


Figure 16: Actual WEKA J48 tree for CGPA using only the 12 questionnaire inputs.

The main CGPA tree starts with recent SGPA. For students in the highest recent-SGPA band, outside commitments and desired-department match provide the remaining splits. The resulting six-leaf tree reaches 49.77% on the fixed test set. This single-split result complements the more stable 10-fold Random Forest result; it does not replace it.

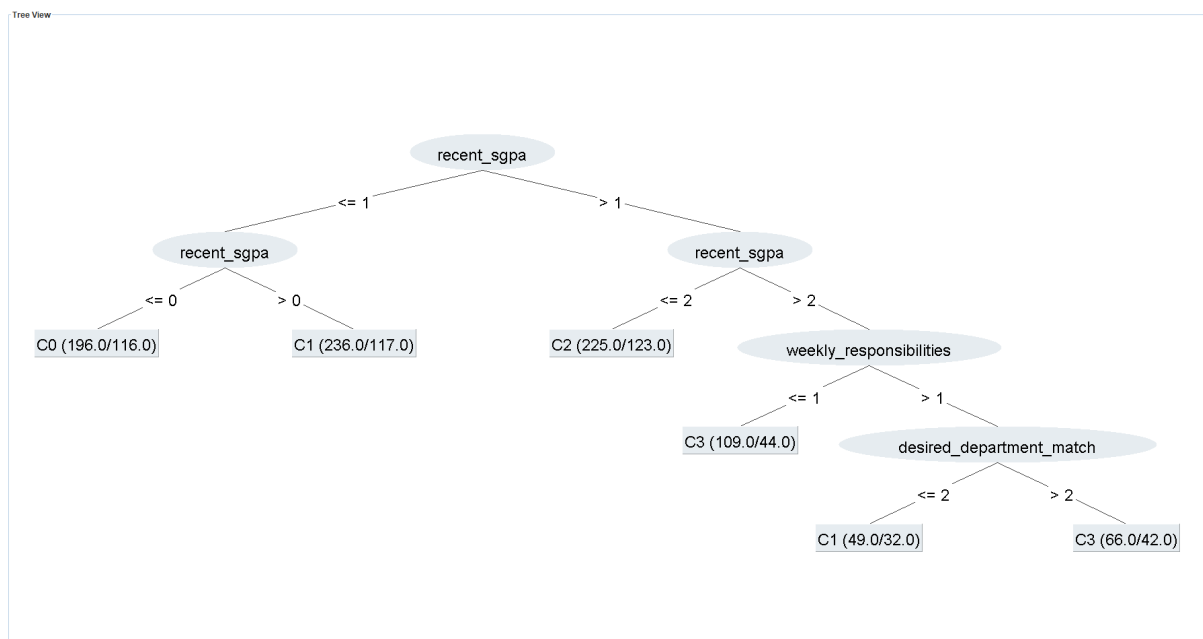


Figure 17: Actual WEKA J48 tree for the main CGPA task. Recent SGPA forms the first split.

9.4 Main-tree confusion matrix and ordered errors

The main J48 tree correctly classifies 110 of 221 test students. Rows in Figure 18 are actual groups and columns are predicted groups. Seventy errors are one adjacent band away and 41 are two or three bands away.

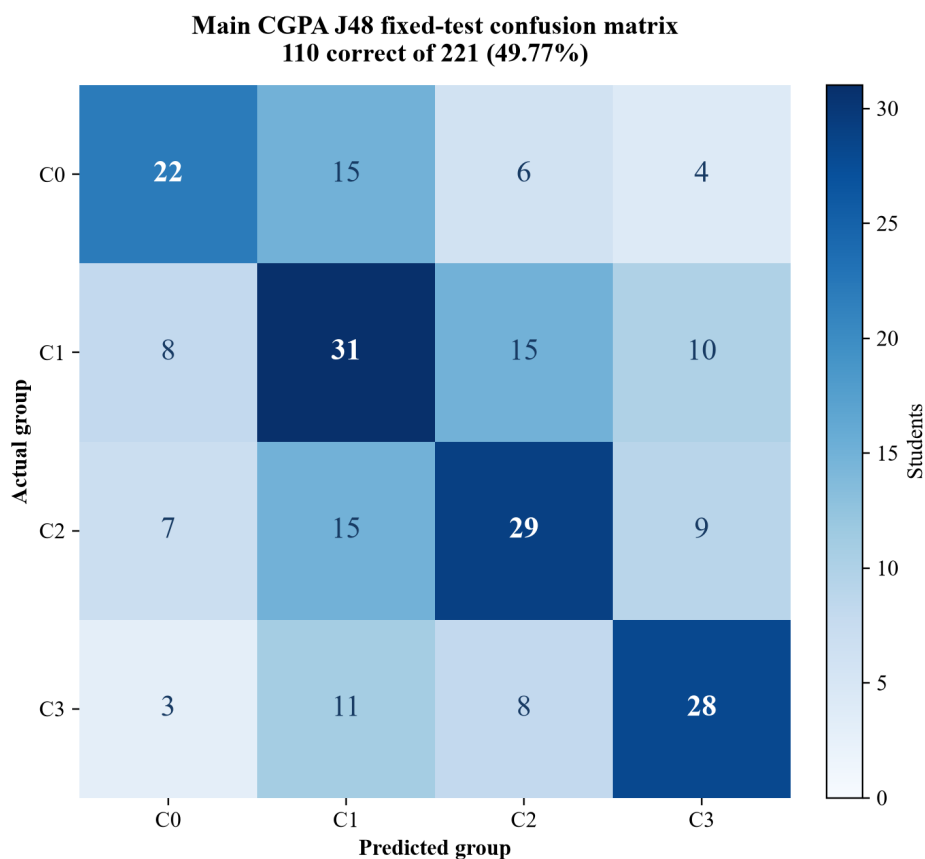


Figure 18: Fixed-test confusion matrix for the main WEKA J48 tree.

Table 10: Ordered error distribution for the 221-student fixed test set.

Absolute class distance	Students	Share (%)
0: exact group	110	49.77
1: adjacent group	70	31.67
2 or 3: farther group	41	18.55
Mean absolute class distance	0.719 groups	

9.5 WEKA output evidence

The repository stores the raw text log, serialized model, textual tree, DOT graph, TreeVisualizer image, and rendered output capture for each J48 target. Figure 19 shows the main-task output generated from the final WEKA 3.8.7 log.

```
WEKA 3.8.7    main    J48 -C 0.25 -M 40

=== TEST ===

Correctly Classified Instances      110          49.7738 %
Incorrectly Classified Instances    111          50.2262 %
Kappa statistic                    0.3246
Mean absolute error                 0.3339
Root mean squared error             0.4075
Relative absolute error             89.529 %
Root relative squared error         94.3535 %
Total Number of Instances          221

=== Detailed Accuracy By Class ===

      TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
      0.468    0.103    0.550    0.468    0.506      0.388    0.759    0.428    C0
      0.484    0.261    0.431    0.484    0.456      0.216    0.628    0.385    C1
      0.483    0.180    0.500    0.483    0.492      0.306    0.657    0.406    C2
      0.560    0.135    0.549    0.560    0.554      0.423    0.711    0.469    C3
Weighted Avg.    0.498    0.177    0.502    0.498    0.498      0.324    0.683    0.419

=== Confusion Matrix ===
  a  b  c  d  <-- classified as
22 15  6  4 | a = C0
 8 31 15 10 | b = C1
 7 15 29  9 | c = C2
 3 11  8 28 | d = C3
```

Figure 19: Rendered WEKA output for the main CGPA J48 run, including settings, accuracy, and confusion matrix.

9.6 Comparison with previous studies

Table 11: Related-work comparison. Accuracy values use different data and targets and are not directly comparable.

Study	Population or target	Models	Reported result and relation
Ahmed (2024) [1]	Student-performance classification	K-means, SVM, Decision Tree, Naive Bayes, KNN	SVM reported 96% after tuning. The current work emphasises a merged engineering survey and target-specific validation.
Maruf et al. (2024) [2]	Bangladeshi SSC and HSC results	Tuned models, stacking, SHAP	Stacking reported 82.23% for SSC and 86.89% for HSC. The target and population differ from four-band university CGPA.
Albonny and Duru (2026) [3]	More than 8,000 secondary-school students; course-grade regression	Linear Regression, Decision Tree, Random Forest, DNN	Random Forest reported $R^2 = 93.3$. The current project uses ordinal classification and university survey data.
This project	1,102 eligible engineering students; CGPA C0–C3	WEKA J48, Random Forest, SMO, Logistic, Naive Bayes, ZeroR	Random Forest reaches 43.92% under 10-fold validation; the readable J48 reaches 49.77% on one fixed 221-row test set.

10 Discussion

Group-level association and individual prediction answer different questions. Attendance, outside commitments, and desired-department match vary modestly across CGPA groups, but the WEKA questionnaire-only models remain close to the majority baseline. The best questionnaire-only 10-fold result is 30.04% compared with 29.04% for ZeroR. Students in different CGPA groups often provide similar questionnaire answers, so small distribution shifts do not create clean four-class boundaries.

Recent SGPA carries a more direct summary of academic history. When it is added to the main CGPA inputs, Random Forest reaches 43.92% in 10-fold cross-validation. The readable J48 tree places recent

SGPA at the root and uses outside commitments and desired-department match only within the highest recent-SGPA branch. That tree reaches 49.77% on the one fixed test split. The cross-validation and fixed-split values use different model settings and should not be ranked as if they were the same experiment.

The fixed split also gives the questionnaire-only tree a small positive change from baseline, while the 10-fold comparison shows only a one-point best-model change. This variation reinforces the need to treat individual branches and single-split scores as exploratory. The stored confusion matrices, models, tree graphs, and raw WEKA logs make each reported value auditable.

The university imbalance limits how far the findings can generalise. The one-time self-reported design prevents causal conclusions. Questionnaire imputation was fitted before evaluation in the historical cleaning pipeline, so limited preprocessing leakage may make the scores optimistic. The results support language such as “associated with,” “more common among,” and “helps prediction.” They do not support claims that a questionnaire answer causes a grade or guarantees a student’s outcome.

Future data collection should prioritise verified grades, course-level results, repeated observations of the same students, objective activity indicators, and better-balanced university sampling. A future evaluation should fit every imputation step inside each training fold before using the system for any decision beyond classroom demonstration.

11 Conclusion

Dataset. Two related Google Forms were harmonised into a documented dataset of 1,108 engineering students from four universities. The workflow preserves raw files, mappings, cleaned tables, figures, ARFF datasets, and audit logs.

Data understanding. The survey responses vary across all answer options. Desired-department match, attendance, and outside commitments show the clearest corrected CGPA associations, but the effect sizes remain small.

WEKA analysis. Every final model experiment uses the same 1,102 eligible records. Questionnaire-only prediction is weak: the best 10-fold result reaches 30.04% against a 29.04% CGPA baseline. For the main CGPA task, Random Forest reaches 43.92% in 10-fold validation. The readable fixed-split J48 tree reaches 49.77% on 221 test students and relies first on recent SGPA. The exact WEKA trees, models, logs, and confusion matrices are included with the project.

The findings describe associations and predictive patterns in the collected survey data. They do not establish causal effects, and the stated sampling and preprocessing limitations remain part of every result.

12 Ethics, authorship, and declarations

12.1 Ethics and privacy

The working dataset omits direct identifiers from modelling, and the presentation assets avoid exposing individual responses. The analysis reports aggregated counts, distributions, and model results. It must not be used for grading, discipline, scholarships, or other decisions about individual students.

TODO: Insert the exact statement that reflects the course survey’s actual consent and institutional ethics procedure. Do not add an approval number or consent claim unless the team can verify it.

12.2 CRediT author statement

TODO: Confirm each team member’s actual roles before submission, then list only verified contributions such as conceptualisation, data collection, methodology, software, validation, visualisation, writing, and presentation.

12.3 Acknowledgements and funding

TODO: Add verified acknowledgements and the accurate funding statement. If the work received no external funding, confirm that fact before inserting the standard no-funding declaration.

12.4 Competing interests

TODO: Insert the team’s confirmed competing-interests declaration.

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A Questionnaire and data dictionary summary

The complete, bilingual questionnaire and detailed coding dictionary remain in the project package. Table 12 provides the modelling variables used in the teacher-facing experiments.

Table 12: Teacher-facing modelling data dictionary.

Feature	Meaning	Type	Missing-value treatment
attendance	Class attendance band	Ordinal	Modal response within source when blank
weekly_study_time	Self-study hours per week	Ordinal	Modal response within source when blank
study_style	Regularity of study	Ordinal	Harmonised label, then within-source fill
topic_clarity	Topic clarity after class	Ordinal	Within-source fill
sleep_duration	Sleep on a normal night	Ordinal	Within-source fill
stress_frequency	Stress blocking study	Ordinal	Within-source fill
weekly_responsibilities	Tuition, job, club hours	Ordinal	Blank mapped to None when supported by form evidence
distraction_frequency	Digital distraction	Ordinal	Within-source fill
study_environment	Suitability of living place	Ordinal	Within-source fill
support_level	Family or friend support	Ordinal	Within-source fill
desired_department_match	Match with preferred department	Ordinal	Within-source fill
routine_manageability	Semester routine management	Ordinal	Within-source fill
recent_sgpa	Most recent SGPA group	Ordinal band	Six affected rows excluded from tree experiments
cgpa	Cumulative CGPA group	Ordered target	Records without usable target removed

B Full model result tables

Table 13: Final WEKA 10-fold cross-validation accuracy on 1,102 eligible students, seed 1.

Model	Recent SGPA	Questionnaire CGPA	Main CGPA
J48	30.13	26.68	38.29
Random Forest	29.76	29.40	43.92
SMO	25.95	27.86	37.11
Logistic	24.95	29.58	36.84
Naive Bayes	26.68	30.04	38.11
ZeroR	26.86	29.04	29.04

C Reproducibility checklist

- Current merged pipeline: `notebooks/02_merged_cleaning_eda_modeling.ipynb` and `notebooks/pipeline_v2_source.py`.
- Teacher-facing analysis: `notebooks/03_teacher_focused_analysis.ipynb`.
- Descriptive snapshot: 1,108 records. All model experiments: 1,102 records after excluding six target-conditioned SGPA imputations.
- Final WEKA evidence: nine ARFF files, 18 cross-validation logs, three fixed-split J48 logs, serialized models, exact trees, and audit files under `weka-final/evidence/`.
- Model comparison: stratified 10-fold cross-validation with seed 1. Explanatory J48: the existing CGPA-stratified 881/221 split with seed 42, reused for all targets.
- Cross-validation J48 uses `-C 0.25 -M 2`; explanatory J48 uses `-C 0.25 -M 40`.
- Run the scripts in `weka-final/scripts/` to rebuild and audit the final evidence package.